

QUESTIONS 8 / 8 completed

Logger

## LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class

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Java

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```
2 public static void main(String args[]){
3     Logger log=new Log4jLogger();
4     log.debug("This is a debug message");
5     log.info("This is a info message");
6     log.warning("This is a warning message");
7     log.error("This is a error message");
8 }
9 }
10 interface Logger {
11     void debug(String message);
12     void info(String message);
13     void warning(String message);
14     void error(String message);
15 }
16 class Log4jLogger implements Logger {
17
18     @Override
19     public void debug(String message) {
20         System.out.println("DEBUG: "+message);
21     }
22 }
```

INPUT

Your INPUT goes here!

OUTPUT

```
DEBUG: This is a debug message
INFO: This is a info message
WARNING: This is a warning message
ERROR: This is a error message
```

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Character

## CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of

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```
1 public class Main {
2     public static void main(String[] args) {
3         Character hero = new Hero("Arthur", 100);
4         Character villain = new Villain("Doctor Doom", 120);
5         Character monster = new Monster("Orc", 80);
6
7         hero.move();
8         hero.attack();
9         hero.takeDamage(30);
10
11         villain.move();
12         villain.attack();
13         villain.takeDamage(50);
14
15         monster.move();
16         monster.attack();
17         monster.takeDamage(20);
18     }
19 }
20
21 }
```

INPUT

Your INPUT goes here!

OUTPUT

```
Arthur the Hero is moving swiftly.
Arthur the Hero attacks with a sword!
Arthur the Hero takes 30 damage. Health left: 70
Doctor Doom the Villain is creeping in the shadows.
```

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Sort ▾

## SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

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Java ▾

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```
1 import java.util.*;
2 class Sort{
3     public static void main(String args[]){
4         TreeSet<Integer> ts= new TreeSet<>();
5         Scanner sc= new Scanner(System.in);
6         System.out.println("how many elements do you need to sort");
7         int n=sc.nextInt();
8         for(int i=0;i<n;i++){
9             ts.add(sc.nextInt());
10        }
11        System.out.println(ts);
12    }
13 }
```

INPUT

```
4
5
6
3
4
```

OUTPUT

```
how many elements do you need to sort
[3, 4, 5, 6]
```

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HashSet ▾

## HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

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Java ▾

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```
1 import java.util.*;
2 public class hset{
3     public static void main(String args[]){
4         Set<String> h=new HashSet<>();
5         Scanner sc=new Scanner(System.in);
6         System.out.println("Enter how many strings do u want to enter:");
7         int n=sc.nextInt();
8         for(int i=0;i<n;i++){
9             h.add(sc.next());
10        }
11        System.out.println(h);
12        System.out.println("Enter which string do you want to remove: ");
13        h.remove(sc.next());
14        System.out.println(h);
15    }
16
17 }
```

INPUT

```
4
hello
world
java
program
world
```

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TreeMap

## TREETMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

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Java

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```
1 import java.util.*;
2 public class tm{
3     public static void main(String args[]){
4         Scanner sc=new Scanner(System.in);
5         Map<Integer,String>m=new TreeMap<>();
6         m.put(1,"joe");
7         m.put(2,"guna");
8         m.put(3,"suba");
9         for (Map.Entry<Integer,String>e:m.entrySet()){
10             System.out.println(e.getKey());
11         }
12     }
13 }
```

OUTPUT

```
1
2
3
```

INPUT

Your INPUT goes here!

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PriorityQueue

## PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

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Java

Run

Save

```
1 import java.util.*;
2 class pq{
3     public static void main(String args[]){
4         PriorityQueue<Integer> p=new PriorityQueue<>();
5         Scanner sc= new Scanner(System.in);
6         System.out.println("Enter how many numbers : ");
7         int n=sc.nextInt();
8         for(int i=0;i<n;i++){
9             p.add(sc.nextInt());
10        }
11        System.out.println(p);
12    }
13 }
14 }
```

OUTPUT

```
Enter how many numbers :
[1, 3, 2, 4, 6]
```

INPUT

```
5
2
3
1
4
6
```

LinkedListHashMap

## LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

```
1 import java.util.LinkedHashMap;  
2 import java.util.Map;  
3  
4 public class Main {  
5     public static void main(String[] args) {  
6         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();  
7  
8         map.put("Apple", 10);  
9         map.put("Banana", 20);  
10        map.put("Cherry", 30);  
11        map.put("Date", 40);  
12  
13        System.out.println("Keys in insertion order:");  
14        for (String key : map.keySet()) {  
15            System.out.println(key);  
16        }  
17    }  
18 }
```

## OUTPUT

```
Keys in insertion order:  
Apple  
Banana  
Cherry  
Date
```

Your INPUT goes  
here!

Max-Min

## MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

```
1 import java.util.*;  
2 class maxmin{  
3     public static void main(String args[]){  
4         Scanner sc= new Scanner(System.in);  
5         PriorityQueue<Integer> p=new PriorityQueue<>();  
6         System.out.println("enter IN how many numbers do you want to find the max and min:  
7         int n=sc.nextInt();  
8         for (int i=0;i<n;i++){  
9             p.add(sc.nextInt());  
10        }  
11        int i=0;  
12        Object arr[]=new Object[n];  
13        while (!p.isEmpty()){  
14            arr[i]=p.poll();  
15            i++;  
16        }  
17  
18        System.out.println("min : "+arr[0]);  
19        System.out.println("max : "+arr[n-1]);  
20  
21
```

## OUTPUT

```
enter IN how many numbers do you want to find the max and min:  
min : 1  
max : 4
```

4  
4  
2  
1  
3